

Exploring Virtual Reality in Exposure Therapy for Sensory Food Aversion

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ABSTRACT

This paper presents research on how to use Virtual Reality with gamified exercises in a therapeutic context with children, focusing on the particular case of warning sensations triggered by sensory properties of food (sensory food aversion). To achieve this goal, we developed a tool featuring several food exposure challenges for patients to use. In the gamified system, the child explores a virtual environment while facing the food they have a problem with. These environments present tasks that resemble typical interactions performed in the real world to develop accommodation. The therapist also has an external system to control the system from outside. In addition, the system sends data collected during the session for the therapist to analyze. We researched how to keep a child engaged in therapeutic tasks and how a child perceives virtual interaction interfaces. The results suggest our system kept the users engaged. Moreover, data show a tendency for the users' results (ease of use, presence, and performance) to remain the same when using controllers or hand tracking. The preliminary results are encouraging and allow us to apply the current system to a wider audience.

CCS CONCEPTS

• Applied computing → Health informatics; • Human-centered computing → Virtual reality; User studies.

KEYWORDS

Virtual Reality, Gamification, Exposure Therapy, Children, Sensory Food Aversion, Digital Health, 3D Environments

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1 INTRODUCTION

Disabilities have existed for a long time, and therapy has been used to help patients improve their quality of life and lessen their difficulties. These impairments include physical difficulties, such as walking or manipulating objects, and psychological impairments, such as low communication skills. During therapy sessions, patients are taught again how to use their muscles and do specific tasks to strengthen them in case of physical problems or learning how to overcome their fear and difficulties in mental cases. However, therapy sessions can be repetitive and thus tiring, leading to a lack of interest in patients. Moreover, the scenarios that can be presented are limited to the physical space available, which is usually the same fixed room. This makes it harder for therapists to create a richer space and customized exercises for each patient.

Adding the use of technology to therapy exercises can have several benefits. One is increased motivation, as using a gadget or an application is often much more interesting than doing something similar without them. Using Virtual Reality (VR) further heightens these benefits since it is a more interesting technology, which also gives a greater feeling of accomplishment when used. VR also has the advantage of allowing the creation of a safe and controlled space where a wide variety of aspects can be tested without leaving the same physical space. A strong point of virtual environments has always been the ability to customize and tailor exercises and games to the patient's needs while creating a compelling, engaging environment. The virtual simulation of therapy exercises presents several challenges and opportunities.

With the current research, we intend to explore how we can use VR to enrich gamified tasks with therapeutic goals that should have high engagement from the user, and that can be adjusted to the patient's needs to produce positive changes to tackle their difficulties. The VR solution applies exposure therapy, which is a type of cognitive behavioural therapy. We focused our study on addressing problems related to food's sensory properties triggering warning sensations in children. Sensory food aversions (SFA) occur when children strongly resist or refuse to eat specific foods due to their sensory processing [Gulácsi et al. 2023]. This reaction can be triggered by the food's appearance, smell, texture, or taste, leading to a negative response. On top of building a solution to tackle SFA, we were focused on answering two research questions: 1) how can a gamified VR therapeutic system keep the child engaged? 2) how does a child perceive the different objects and interactions provided

by the virtual environment, which are trying to represent the real ones?

This paper has the following structure: Section 2 presents concepts and related work. Sections 3 and 4 present the design and implementation of our solution called Pikita VR Quest. Section 5 shows the results of our user study and their discussion. Section 6 concludes the paper by summarizing the research but also presenting future work.

2 RELATED WORK

This section presents and discusses research on the use of VR in the therapeutic field. It is divided into four main topics: two related to physical exercises, immersive exergames and redirection techniques, and two related to cognitive exercises, concept association and exposure therapy to tackle sensory aversion or even phobia.

2.1 Immersive Exergames

Adding VR to therapeutic exercises can easily provide different immersive environments to further motivate and keep the user engaged [Ambrosino et al. 2020; Campo-Prieto et al. 2021a,b]. Corrêa et al. developed an upper limb motor rehabilitation training system using the Leap Motion Control¹ to track the movements of the user's hands [Correa et al. 2019]. In this study, they employed a game where the participant had to pick up and place apples in the correct fruit basket. It has four stages of play, each with an adjustable level of difficulty that varies depending on the sequence of hands.

Bortone et al. devised two serious games for children focusing on upper limb rehabilitation [Bortone et al. 2017]. The system employed immersive VR scenarios as well as robotic haptic devices to recognize a pinching motion. The Moneybox Game focuses on grasping and reaching activities where the player has to take a golden coin and place it in a floating piggy bank. The Labyrinth game has a sliding token that must be dragged through a maze using the fingertip. Both games also have automated difficulty for the size and quantity of items to be gripped or dragged and the complexity of the trajectories to follow. Moreover, Baron et al. developed an interactive VR-based drawing game with two difficulties [Baron et al. 2021]. This game aimed to connect the dots of an outline sketch of the content using a virtual paintbrush.

All of these games test the reach of the users and present different difficulties, which are useful to keep the user's interest as s/he learns or to be used by patients with various difficulties/disabilities.

2.2 Redirection Techniques

A redirection technique is a mechanism used to manipulate the user's movement by adding discrepancies between the real motion and what is seen in the virtual environment. The method used for this is called self-motion gains, and the three most common techniques are translation gains, rotation gains, and curvature gains [Azmandian et al. 2022]. Shum et al. created a game where participants were required to complete a reaching task with several bimanually interactable item models [Shum et al. 2020]. The scenery and tasks were designed to resemble food preparation.

Another game used a feedback technique known as error augmentation (EA) that has been used to adjust feedback based on the amount of error, typically amplifying the error, keeping the user engaged and motivated to reduce movement errors [Liu et al. 2018]. As Liu et al. conclude, overall, EA significantly improves clinical and kinematic outcomes compared to standard repetitive practice without error modification.

Both [Sror et al. 2019] and [Cho et al. 2022] built systems that support arm movement in the horizontal plane and make use of a VR interactive game and error augmentation where the users have to perform a reaching task. After four weeks of training, the patients showed improved results regarding time, speed, and distance in all directions tested, as well as significant improvements in upper-limb motor recovery characteristics [Cho et al. 2022].

These last three works used EA, a simple but interesting technique that accentuates the error to obtain a compensatory response [Liu et al. 2018]. This technique makes the task harder, which is a way to keep the user's interest, showing positive physical results. The last two games, [Sror et al. 2019] and [Cho et al. 2022], offer environments that are too simple to be able to capture a child's interest. For this, these games need to include changes in how the challenge is presented, giving a small backstory to motivate the user or using better visual components.

2.3 Concept Association

Concept association is a type of brain game that can improve cognitive function regarding memory, solving problems, and thinking critically. Taking part in this kind of game for a few hours a week can produce lasting benefits².

Funabashi et al. created a serious game that takes advantage of serious game and VR concepts to give a pleasant approach to rehabilitating post-stroke patients [Funabashi et al. 2017]. The objective of the game is to match one target image to another that is similar or complementary from a pool of four options.

Both [Chatterjee et al. 2022] and [Caggianese et al. 2018] developed cognitive training systems, allowing patients to do daily activities. For Chatterjee et al., the developed scenarios include a kitchen scene where one of the tasks was to make a toast and a cafe scene, in which one of the tasks was paying for a meal. Caggianese et al.'s system offers many tasks with varying levels of difficulty with tasks, such as cooking activities.

The games presented in this subsection focus on improving the cognitive difficulties of their players, focusing on memory and reasoning. Regarding the last two games, if they were to be used by children, they would need adaptation by adjusting the content and context of the tasks to make them more familiar to children.

2.4 Exposure Therapy and VR

Behavioral therapy is a broad term that encompasses various techniques designed to modify maladaptive behaviours [Fordham et al. 2021]. The primary goal of behavioral therapy is to reinforce desirable behaviors and eliminate unwanted ones. Exposure therapy is a specific type of behavioral therapy that is particularly effective for treating anxiety disorders, including phobias, panic disorder, social

¹ Ultraleap website, tracking devices company <https://www.ultraleap.com/product/leap-motion-controller/>. Last Accessed: 2023

² St. Luke's Health, health care provider <https://www.stlukeshealth.org/resources/brain-games-improve-cognitive-ability>. Last Accessed: 2023

anxiety disorder, and post-traumatic stress disorder (PTSD). The core idea is to expose the patient to the feared object or context without any danger to help them overcome their anxiety [Wendt et al. 2018]. Implementing a phobic element in the VR environment can provide virtual stimuli under a safer environment [North et al. 1998].

Boyd et al. created a multi-user game produced through collaborative design that delivers real-time input on proximity to another user, the user's speaking volume, and time spent conversing [Boyd et al. 2018]. A solution for acrophobia therapy in an urban environment was developed by Anton et al., where a single-player VR serious game exposes people to heights in the urban environment [Anton et al. 2020]. The goal is to reach the top of the tower while overcoming their fear of heights, presenting elements of gamification in the form of mathematical quizzes to ease the exposure.

Go et al. built a system that included exposure therapy with a VR application [Go et al. 2020]. Each level had the player completing objectives and gathering stars while surrounded by animals or locations they were afraid of. The game was developed to treat five different phobias and, for each of the phobias, several levels were created, each with different exposure levels. The level designed for acrophobia is similar to the previously discussed game in [Anton et al. 2020], except for the part of the quizzes.

All of these systems took the approach of letting the player get used to their fear at their own pace in a controlled environment provided by VR, which can safely be left at any moment the player feels uncomfortable.

3 PIKITA VR QUEST FOR EXPOSURE THERAPY

In this section, we start by presenting the steps followed to develop our solution called Pikita VR Quest. After this, we present the main concepts of the gamified VR system and the interactions developed for the therapist. Finally, we present the hardware devices and software tools used to develop the VR application as well as how these two components communicate.

3.1 Design Process and Concept

The development of the system followed as much as possible the design thinking methodology, which is composed of five main stages [Roddy and Polfuss 2020]: Empathize, where we seek the users' perspectives and frustrations; Define, where we identify those perspectives and frustrations to define the problem; Ideate, where we brainstorm ideas about what was identified; Prototype, where we create an observable version of the solution; and Test, where we have the users test the solution and get feedback.

During the development of the system, we had several meetings and workshops with the therapists we were working with (who acted as our clients in place of the children) to mould our program to their and the patient's desires and needs, refine some features, add new ones, or scrape others. This led to the system's core being complete, although not refined, from early on. From this, each week that passed, the system suffered several minor changes that slowly allowed it to reach its final stage.

After initial workshops and interviews with the therapists, we put together the following set of principles that our system would

need to follow: **Gamification, Simple tasks, Comfort, Motivation, Data gathering, Connection to real world, Therapist configuration, Therapist interaction with real-time changes, External visualization.** Additionally, to answer our second research question, we needed features that would allow us to create experiments for perception tests.

With the principles and requirements set, we built a case study that included them. The case study had at its base the creation of a gamified version of tasks that would resemble exercises typically done in exposure therapy to tackle sensory food aversion, which can progress to food phobia if it is extreme, specifically within the context of Avoidant/Restrictive Food Intake Disorder (ARFID). Individuals with ARFID often exhibit extreme sensitivity to the taste, texture, smell, or appearance of food [Kreipe and Palomaki 2012]. This heightened sensitivity can cause significant distress and lead to avoidance of various foods. Over time, the anxiety and fear associated with negative sensory experiences can evolve into a food phobia [Fonseca et al. 2024]. This phobia is characterized by an irrational and persistent fear of certain foods or the act of eating.

In these exposure therapy sessions, the therapist can progressively introduce tougher interactions with the object of aversion to create habituation to those interactions gradually. Afterwards, we set the experimental setup considering the requirements and the case study. The patient will play standing up in an area without obstacles that will allow them to move freely. Inside the game, the patient will be presented with a spacious room where they will face the challenges. We decided to develop tasks with different difficulties and work on different aspects. While the patient is playing, the system will collect information about the session. The therapists will also have their part of the system in the form of a web application where they can configure and interact with the VR environment in real time while seeing what the patient is doing. Therefore, the therapist will accompany the patient during the different tasks that are presented while indirectly controlling the system.

Ideally, we would have tested the system with the target audience during development. Yet, we did not do it because we considered that the system needed to have a higher refinement level before going into their hands. Therefore, we tested the system, for the most part, with only adults with no problems with food. By doing this, we could create a solid and usable base from which to work. After this was set, we progressively lowered the age of the users that were testing, approaching our target group's age. Following the same rationale would allow us to, at each time, find possible problems that could emerge, creating a more robust baseline. This would be especially important when we test with children, but without the food problems, as we can first establish which issues come from a usability standpoint and which come from a difficulty posed by the degree of aversion they have from the object they are facing.

The major part of our system is the VR application that we built to satisfy the requirements established previously. The main components of our Pikita VR Quest can be seen in Figure 1. As shown in it, we have several environments used to test the warning sensations triggered in the patients. Other than these, we also built challenging

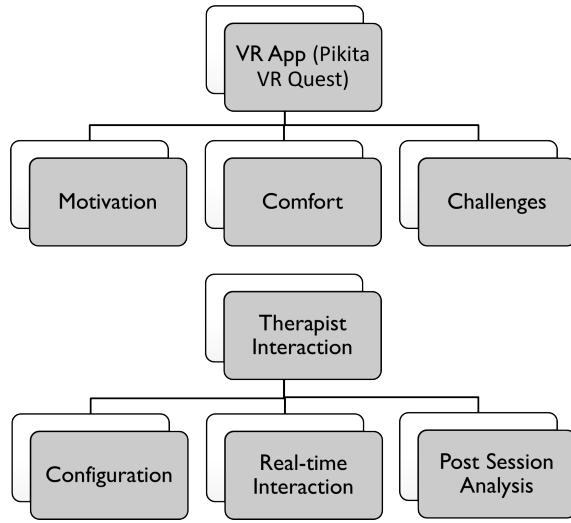


Figure 1: Diagram comprising the main concepts that make up the VR application and the Therapist's system

environments to help them overcome their problem. These challenges were gamified to keep the children engaged, and a set of motivational features was added to make them more attractive. By completing the challenges, the players can earn points and unlock an interaction with a reward object. These challenges progressively become more challenging when testing different interactions by making the patient closer to the virtual problematic objects and even interacting with the real version of those objects in the final challenge. The base environment for these challenges can be seen in Figure 2, with the others having the same or similar displays. Finally, we added a set of comfort features used to help the patients during the eventually stressful gameplay.

For all of the challenges, we can set the size of the play area, represented by the area surrounded by the fence. This size should reflect the size of the physical area available as every object's position and other distance-related variables will be adjusted according to this.

3.2 Motivation

We needed to keep our users engaged, as is the case for any game, so we implemented a variety of features designed to motivate them to play. These features are a point system (showing the current amount of points, showing the number of available points and their decrease); displaying the **challenge-timer** for each object; **confetti** falling, display of the **points received** and the sound produced by it when completing a task, as seen in the bottom left of Figure 3; and interaction with a **star**, and the sound produced by it after all tasks are completed.

3.3 Comfort

Being faced with a stressful situation can increase the patient's anxiety. To help with this factor, we decided to include a few ways to deal with it in our system, which are the following: **Comfort character**, seen in Figure 2, that would accompany the patient



Figure 2: Image of the base environment. 1- Play area; 2- Comfort character; 3- Problems objects; 4- Hands of the player; 5- Current points; 6- Available points

throughout the challenge environments; **Safe Space**, where the patients could go if their anxiety levels were too high; which contained **Comfort objects** (bottom right of Figure 3).

The Comfort character was animated to perform movements similar to the ones needed to complete each challenge. Doing this showed the patients that it was a safe interaction. For the Safe Space, we decided to create a happy environment that would not have any problematic objects and would instead have Comfort objects. To let the players know that they are approaching the zone's boundaries, we incorporated a change in the screen's colour, which would darken as the limits of the play area were reached. At a certain point, the player would be moved to the Safe Space and they would need to return to the center of the play area in order to return to the challenge.

3.4 Challenges

3.4.1 Reaction Challenges (Passive). These challenges are used by the therapists to see how the children react to the objects (e.g., if they look away, if they are afraid), which can be used for an initial assessment of the warning sensations triggered by the objects. Additionally, these challenges are the most basic environments, which do not require direct interaction with the objects. We decided to create five levels. Although there were no direct interactions with the objects, the patient still held the controllers for the system to gather data. The summarized information on the challenges, including their designation, description, and justification for creation, can be seen in the first half of Table 1.

3.4.2 Interaction Challenges (Active). As explained before, the challenges are simple in themselves to be easy enough for a child to understand. This was done so that the challenge itself would not pose additional stress to the patient who is already facing an uncomfortable situation. The summarized information of the challenges, including their designation, description, and justification for creation, can be seen in the second half of Table 1, and the environment can be seen in the top left and top right of Figure 3.

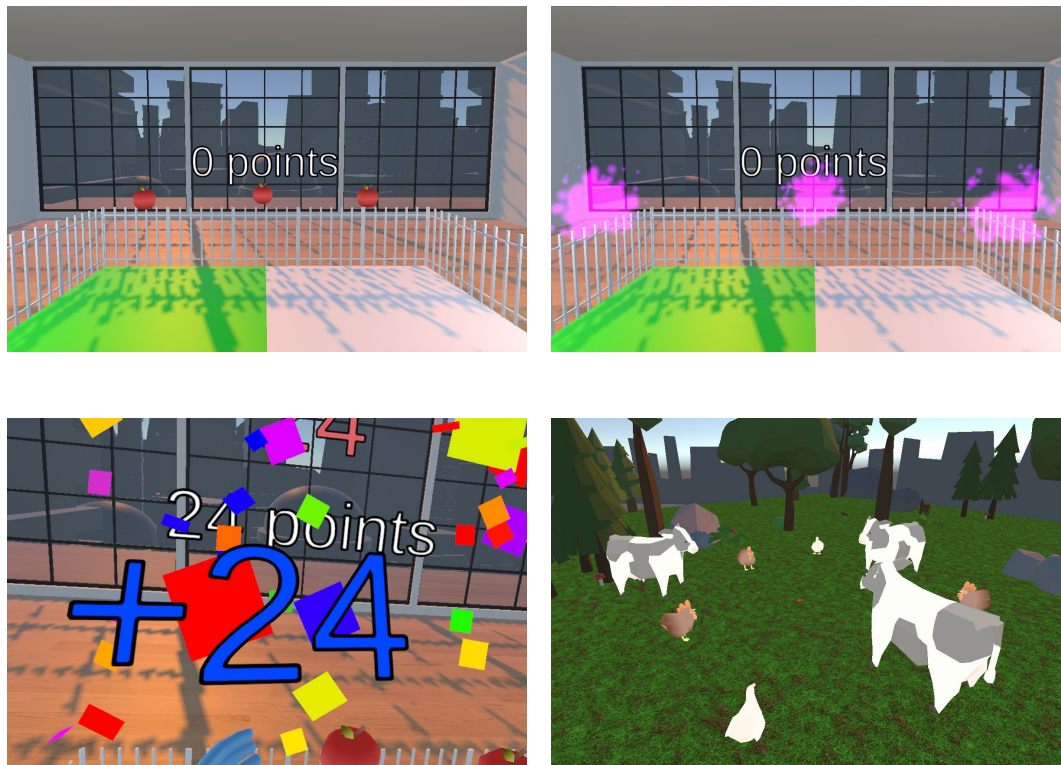


Figure 3: Top left: Exposure to problem object (Interaction Challenges 1 - 4); Top right: Clouds to hide aversion real objects (Interaction Challenge 5); Bottom left: Confetti falling and receiving points when the task is completed; Bottom right: Safe Space and Comfort objects

3.5 Therapist Interaction

We built the therapists' interfaces to satisfy their need to control an array of different options to which they do not have access in commercial games. This control will allow them to manipulate and change the game according to each patient's needs.

As such, therapist interaction is a crucial aspect of the game flow of our system. Our system allows the therapist to change some configuration options before the session and control some aspects of the system while the patient is playing. Given that the VR device we used was the Meta Quest 2, which is a standalone device, this interaction would need to be triggered either directly using the device or externally by another application. As such, two different interfaces were created for the therapist. One was done inside the game, and the other was a web application, PLAY [Antunes and Madeira 2022], whose objective is to give the therapists a single place from where they can control several games as well as have information regarding each patient. It is also worth noting that the requirement of external visualization was accomplished by using the cast function on the Meta device, which can be seen by logging in to the website or mobile app.

The interfaces were divided into three parts, as seen in Figure 1: one for configuration, another for real-time interaction and, finally, one for post-session analysis, which is made available by PLAY with

the data sent from the VR application. Configuration was available in both solutions, while real-time interaction and post-session analysis were only in PLAY. The configuration consisted of adjusting the size of the **play area**, selecting the **problem** and **comfort** objects, selecting other features to be enabled, and choosing the maximum amount of **points** for each challenge. The real-time interaction consisted of changing **environments** and performing **actions**, such as animating the Comfort character or starting the challenge.

3.6 Technology

The computer used to develop the system had an NVIDIA GeForce GTX 1050 Ti graphics card and an Intel(R) Core(TM) i7-10700F CPU with 2.90 GHz and 16 GB RAM. Along with this computer, the main VR device used for development was the Meta Quest 2. In the experiment, two different hardware devices were used. One of them was the same VR device used for development. The other device was a PC that ran the therapist application. The PC had an Intel(R) UHD Graphics 620 graphics card and an Intel(R) Core(TM) i5-8250U CPU with 1.60GHz and 8 GB RAM. Our system was developed in Unity, a game engine that provides an API to work with several devices.

Table 1: Reaction Test and Interaction Challenges summary

Challenge Environment	Task Description	Exposure Justification
Reaction (Passive Tests)		
1. Idea of the FA object (represents the food of aversion)	The user sees a ball with the colour of the FA object, sees and ears the name of the FA object	Make the user think about the FA object without seeing it
2. Seeing less FA object (one direction)	The user sees three copies of the FA object next to each other	Expose the user to the FA object while having the objects restricted in space
3. Seeing more FA objects (surrounding)	The user sees eight copies of the FA object scattered around them	Expose the user to the FA object without the objects restricted in space
4. Moving FA objects	The user sees four copies of the FA object on top of a table; after 10 seconds, the FA objects start rolling towards the user	Expose the user to the FA object while adding movement to them
5. FA Objects falling from the ceiling	The user sees copies of the FA object falling from the ceiling	Expose the user to the FA object while exaggerating its presence
Interaction (Active Tests)		
1. Proximity	The user has to be at a certain maximum distance from the FA object for a certain amount of time	Make the user get used to being closer to the FA object
2. Looking	The user has to look at the FA object for a certain amount of time	Make the user get used to looking at the FA object
3. Grabbing	The user has to grab and hold the FA object for a certain amount of time	Make the user get used to grabbing and holding the FA object
4. Eating	The user has to grab the FA object and bring it close to the mouth	Make the user get used to the movement of bringing the FA object to the mouth
5. Hidden Real FA Objects	The user interacts with real FA objects presented by the therapist who also controls the challenge completion	Have the user interact with the real FA object

Figure 4 presents a system architecture diagram. We can see the components presented in the previous sections and some notable packages we used, all from Unity. These were the XR Interaction Toolkit and XR Hands for the development of VR interactions and access to the hand-tracking data. Communication with PLAY comprises two protocols: WebSockets, which is used for bidirectional real-time communication, and HTTP, which is used for all other calls.

4 USER STUDY

In this section, we start by presenting the different interaction interfaces studied as well as the characteristics of the participants. After this, we present the analysis of the results obtained in the tasks as well as their discussion. Following this, we present guidelines and limitations. Finally, we present feedback obtained after discussing our final progress with the therapists team.

4.1 Perception Test Interaction Interfaces

One of our objectives was to investigate the difference in perception of the users in a virtual environment compared to the real world. In this user study, we decided to explore this with different **interaction interfaces**. The interaction interfaces chosen were:

- C - Using and seeing **controllers**
- H - Using and seeing **hands**
- M - **Mixed**, using controllers while seeing hands

4.2 Participants Characteristics

Our initial goal was to perform this study with children who had difficulties dealing with food. Since this is a sensible target audience, initially, we wanted to test the system with children without difficulties. For now, only this testing phase is completed, with the other being left for future work.

21 children (12 male and 9 female) participated in the experiments. All of the participants were students in grades 2 to 4, and their ages ranged from 7 to 10 (mean = 8.76). Out of the 21 participants, only two had prior experience with a VR device. Regarding the experience with video games, ten participants play around three times per week, seven said they play only once, and the remaining four less than once a week. None of the participants said that they had any type of deep problems with food, and none said they knew this was an existing problem.

4.3 Tasks Results

We had the participants take part in two tasks where they played the corresponding environments and answered questionnaires.

4.3.1 T1 - Interaction perception. This task was done by having the patients perform the same challenge (Grabbing) using the three types of interaction. The data gathered was the presence and ease of use regarding each interaction, measured on a 7-point Likert scale, along with the points received.

Subjective Ease: For this metric, the Friedman test ($\chi^2 = 5.85$, $p = 5.34 \times 10^{-2}$) shows that there is no significant difference between

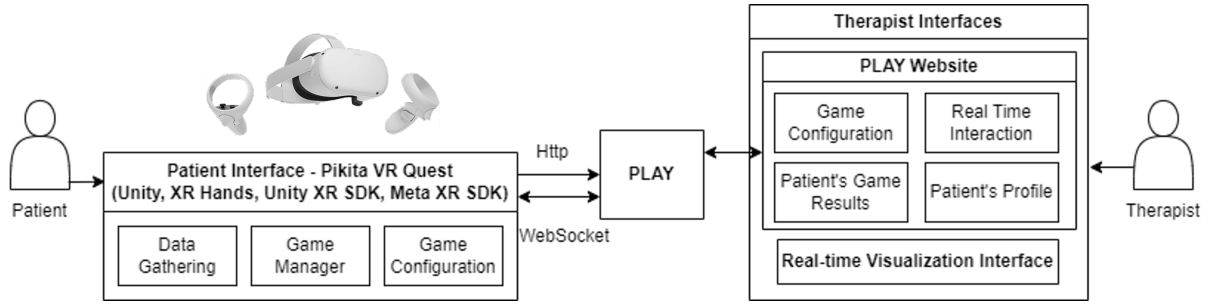


Figure 4: Diagram of the system's architecture

Table 2: Statistical analysis of the presence felt while playing in T1

Interaction	Presence (7-point Likert scale)			Friedman Test $\alpha = 0.05$	
	Median	Q1	Q3	χ^2	p
C - Controller	7	-2	+0	2.92	2.31×10^{-1}
M - Mix	7	-1	+0		
H - Hands	7	-2	+0		

Table 3: Statistical analysis for the performance (points) during the challenges in T1

Interaction	Points			Friedman Test $\alpha = 0.05$	
	Median	Q1	Q3	χ^2	p
C - Controller	66	-6	+3	7.54	2.31×10^{-2}
M - Mix	69	-6	+3		
H - Hands	66	-3	+3		

any pair of the data sets. Since the result was still close to being smaller than 0.05, we applied the Wilcoxon signed-rank test to the three pairs. The results of the tests suggest that while there is no difference between the controller and any of the other two types, there is a significant difference between the mixed and the hand interactions, with $p = 3.31 \times 10^{-2}$.

Subjective Presence: In Table 2 we can see that the result of the Friedman test applied to these data sets ($\chi^2 = 2.92$, $p = 2.31 \times 10^{-1}$) shows that there is no significant difference between any pair of the data sets.

Points: The maximum amount of points received in the challenges was 30. Each time the available points are reduced, they are lowered by 10% of the original value. Given this and that the first reduction takes place three seconds after the start, the second 4.5 seconds after the first, and the third 6.75 seconds after the second, the maximum number of points possible in each challenge is $2 * 27 + 1 * 24 = 78$ as each challenge takes 5 seconds to complete. This result can only be achieved if both hands are used simultaneously; otherwise, the maximum is $27 + 24 + 21 = 72$. The minimum number of points is $3 * 15 = 45$.

Table 4: Motivation Questionnaires results summary

Question	Median	Q1	Q3
1. Help from challenge-timer in food.	6	-4	+1
2. Motivation from confetti and receiving points.	7	-1	+0
3. Motivation from the point system.	7	-0	+0
4. Motivation from seeing current points.	7	-1	+0
5. Motivation from seeing available points decrease.	7	-2	+0
6. Motivation from interacting with the star.	7	-1	+0

In Table 3, we can see that the result of the Friedman test applied to these data sets ($\chi^2 = 7.54$, $p = 2.31 \times 10^{-2}$) shows there is a significant difference between at least one of the pairs of the data sets. By applying the Wilcoxon signed-rank test to the three pairs, we can find a significant difference between the controller and mixed interfaces ($p = 1.67 \times 10^{-2}$). The other two pairs showed no significant difference.

4.3.2 T2 - Regular challenges. In this task, the participants used the system in its normal context. They played the looking, eating, and hidden real FA objects challenges. After that, they answered the following questionnaires on a 7-point Likert scale with help from the researcher: one about the motivation features, another was the System Usability Scale [Brooke 1996; Martins et al. 2015], and the IGroup Presence Questionnaire [Schubert 2003; Vasconcelos-Raposo et al. 2016]. The participants were told to take into account their experience in this task only when answering the questionnaires.

Game Specific Questions: (seen in Table 4) In questions 1 and 5, some participants said that they recognized the feature when shown an image but did not fully understand what it did (e.g., they saw the timer in the object and saw it changing value but did not know what it was representing), with some not even recognizing the feature evaluated in question 5. These participants did not answer this question, which can inflate the result. The results shown in Table 4 suggest that the participants felt motivated by the features tested. The only one that did not achieve this result was the challenge timer.

System Usability Scale: This questionnaire was used to measure the usability of Pikita VR Quest. Some of the questions were asked

Table 5: IGroup presence questionnaire results summary

Question	Median	Q1	Q3
1. I had a sense of "being there".	7	-1	+0
2. I was just perceiving pictures.	2	-1	+3
3. I had a sense of acting in the virtual space.	6	-1	+1
4. I was not aware of my real environment.	7	-1	+0
5. I still paid attention to the real environment.	1	-1	+1
6. I was completely captivated by the virtual world.	7	-1	+0
7. The virtual world seems real.	6	-1	+1
8. The virtual world seems consistent with the real world.	5	-2	+1
9. The virtual world seems more realistic than the real world.	2	-1	+1

a bit differently to make them easier for the children to understand. Overall, the results of all questions were very good, with only one question without a median of the best result. The average SUS score was **90.24**.

IGroup Presence Questionnaire: The questions used from this questionnaire can be seen in Table 5 and were used to evaluate the presence felt by the participants when playing the game. For some of these questions, the higher the result, the better (1, 3, 4, 6, 7, 8, 9), while for the others, the lower the better (2, 5). Some of this questionnaire's questions were removed to decrease confusion among the participants. By observing the results, we can see that, overall, the answers to every question were positive, except for the last one, which we were already expecting, as it would be very extreme. Out of the other 8 questions, the question that stands out as having the worst result is question 8.

4.4 Discussion

Overall, the experiment's biggest obstacle was the fact that we were dealing with children. To talk to them, we had to adjust our vocabulary and explain some concepts in terms that they would understand. This was particularly noticeable during the questionnaires, which, along with the shyness of some children, possibly influenced the results.

In T1, we were expecting a significantly worse result in terms of ease of use and points with the hands interface when compared to the other two, while we were expecting better results in presence. For the ease of use, we can only assume that they felt the game was so easy to play that even a slightly more difficult interface would still be a 7/7. In terms of presence, we are more surprised as we expected the controller interface to perform quite poorly and the hands interface to have a good result. The explanation for these results might be that the participants did not understand the explanation about the term presence or did not pay enough attention to their virtual hands to have them influence the level of presence felt.

In T1, two modes of interaction were played using the same physical controls, only changing the visualization. Yet, the points overall were higher in the mixed interaction, with the statistical tests confirming the significant difference. Having no significant

difference in terms of presence or ease of use, the results suggest that the participants probably learned from the first challenge and applied the knowledge gained in the second challenge. Taking this into account, along with the other results, we can assume that none of the interaction modes stands apart. While this can be seen as a negative conclusion, it can also be viewed as the opposite. Knowing that all modes produced similar results means that being able to select the interaction mode to play with would introduce a difference in play that could produce a bigger interest from the player. Thus, this could also be used as another tool to keep the player motivated and engaged in the game.

In T2, we were quite surprised that the challenge timer did not perform well as we were expecting the participants to easily understand what it represented and to be helpful information by showing the amount of the task completed. We think this might have happened because the participants did not need any confirmation that they were progressing through the task. Regarding the participants who did not see the display of the points available, we think it happened because of the height at which it was shown. Since the participants had to look down to see the objects and up to see the display, they did not notice it. Additionally, this display only appeared when the challenge started, when they were more focused on completing the task, and was often only noticed after completing it.

Our system's score on the system usability scale had a very high value, which can be explained by the fact that it was developed with tasks as simple as possible. Despite this, the result was still somewhat surprising as we did not expect children of that age, who mostly never used VR, to rate it this high. Bangor et. al present the average SUS score for different types of applications [Bangor et al. 2009]. Although there is no representation of virtual reality applications, we can assume the highest shown, which was the graphic user interfaces at 76.2, which is still lower than our score of 90.24. In the presence questionnaire, question number 8 had the least positive result (after 9), which could be related to one observation made by some participants who said that, when holding an apple, the hands would not correctly wrap around it. This was one feature that could have been developed but was deemed unnecessary because we thought the participants would be more focused on the environment and feel more challenged to notice this.

4.5 Guidelines and Limitations

Between the test tasks, we think the ones that performed the best were the environments played in T2, especially in the first two challenges, Looking and Eating. We feel these were the environments that left the participants more at ease as they did not feel the pressure of questions asked after playing each scenario in T1. From all of the motivational features, we felt the one that had the most impact on the performance of the participants, even before considering the result shown in Table 4, was the overall point system. This feature gave some of the participants a clear objective – get the most points possible. Even if they did not understand how the point system worked or were sure how to play, they wanted to complete the tasks as fast as possible. This could have been the case had they assumed it works similarly to other commercial games.

This points system also gave a quantitative measurement of performance, which the participants could compare with each other post-session, therefore taking advantage of their competitiveness to increase engagement with the tasks.

The tasks/sub-tasks we felt had a performance below expected were the questionnaires in T2. The solution to this problem is difficult as it is tied to the mental ability of the children. Having a child answer a questionnaire is already hard as they might not understand the questions. On top of this, it would be even harder to have children do that after playing a game they were playing for fun, not thinking about looking around and paying attention to certain details. The best change we can think of for this problem would be to test the features as separately as possible so one of the features will not distract them from the others. The features we felt had results below our expectations were the object challenge-timer and display of available points for the reasons already explained. Regarding the challenge-timer, some of the participants felt that it did not add much to the gameplay, while others felt it was a worthwhile addition that helped them understand how much of the task had been completed. Given this, we arrive at a stalemate where, on one side, it has good reasons to stay, and on the other, it is just cluttering the vision of the player. Putting that aside, we could also make it more visual with a progress bar instead of a numeric representation of the time. We feel the available points display should be adjusted in one or both of two different ways. The first is to change it to a display of time, which would have an objective that is more natural to recognize. The second is to change the position it appears in to be more visible during the challenge or have it appear from the beginning, which would have the player know it exists and pay more attention to it during the challenge.

Finally, as explained in the previous section, we could implement the correct wrapping of the virtual hands around the objects. Other than this, the hidden real objects challenge would perform better if played in a Mixed Reality setting. This would allow the user to see either the object they were interacting with or a representation of it instead of not seeing anything. This would make the challenge more visually appealing, feel more natural to play, and give a higher degree of immersion.

4.6 Therapists Interview

After performing the user studies previously shown, we met with the two therapists we work with to participate in a technical interview. In this interview, we presented our environments and features, discussed them, and got feedback one final time before starting to make tests with therapists and their patients. This interview was more inclined to put the VR application to the test since that was the one that could potentially be harmful to the patients and/or their therapeutic progress.

Other than qualitatively discussing the challenges, we asked the therapists to give a grade from 1 to 7 on how good they thought the challenge was. The Reaction Test Challenges were generally liked, with the middle 3 receiving a grade of 7. The Idea of the Object and Falling objects were the ones that left our therapists in disagreement, with one feeling that the first was too simple and liked the last, while the other liked the first but felt the last was too intrusive and would have a high impact.

Overall, the Interaction Challenges also had good results, with the first three receiving a grade of 6. This mark did not reach a 7 due to the therapists feeling the challenge-timer could be discarded and instead have the object fading out until disappearing when its task was completed. The Eating challenge received a 5 as the therapists said the way it was presented should be changed, instead of saying "eat the object" we should say "bring the object to your mouth". The Hidden Real Objects challenge was the one with the worst mark (only 4), as the therapists said the portals were too visually intrusive and should be replaced with something more concrete, such as a box or a basket. After discussing the results of our test study with the therapists, they also found the idea of implementing this challenge in a mixed reality setup as an additional challenge. Even with all the issues related, the therapists deemed all the challenges to be at least reasonable to be tested with the patient, except for the Falling Objects challenge, which one of the therapists felt could have a negative impact.

All of the motivation features performed well in their eyes, and we found that motivating the patient would be a welcome addition to our system and the therapy session. The Comfort character had already been a previous suggestion on their part, which they felt was accomplished. Comfort objects, the Space Space, and the screen darkening were also well accepted, with one exception. The darkening should not transition into the Safe Space as this causes too much for the patient to get distracted and try to explore the Safe Space instead of doing the challenges.

5 CONCLUSIONS AND FUTURE WORK

Regarding our objectives, we were able to design a system composed of features that therapists deemed worthy of using during sessions with their patients. The environments developed were able to replicate the exercises done in conventional therapy, and the therapist system successfully gave sufficient control over the VR component. However, it still left them wanting more. From the user studies and therapist interviews, we were able to take away that our motivational features produced good results overall and are always good additions, with the best performing out of them being the existence of a points system. Although, in practice, some of these features did not produce the best results, we now have ideas on how to modify them. Regarding the interaction interfaces' results, although we could not find any significant difference suggesting that one was better than the others, all of them still managed to get good results.

On top of the changes that resulted from our user studies, others need to be addressed to have our system up to the standards of the target audience. Following this, we can begin the tests with them, allowing us to truly test our system and investigate if a virtual object can trigger the same warning sensations as the real one. Accompanying these tests, we will have therapists who have previously participated in an experiment on our user interface. Other functionalities can be added, which include interactions in mixed reality, modular edition of environments, and external addition of new problem and Comfort objects for the game.

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